Communication Lower Bounds for Multi-Party Graph Traversal

Quarter 2 Project Report

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Abstract

A large number of computational problems can be modelled as communication games. For example, a graph traversal problem can be modelled as follows: Two players, Alice and Bob, have subgraphs of a known graph G, and wish to find a path from vertex v to w using only vertices and edges in either player's subgraphs. Communication complexity studies such situations by creating asymptotic bounds on the minimum number of communication necessary to complete these tasks. In this report, we prove communication lower and upper bounds for some graph traversal problems, such as the example above. We also survey how such bounds can be used by algorithm, system, and data structure designers.

Website: https://rybplayer.github.io/capstone/
Repository: https://github.com/rybplayer/DSCCapstone

1 Introduction

1.1 Motivating Communication Complexity

As networks and data grow in complexity and size, it will become increasingly difficult for individual machines to compute functions on such data. For example, suppose we wish to fly from San Diego to Santorini with the least budget. There is no single airline that has both flights out of San Diego and into Santorini. In other words, we will need to consult the flight database of various airlines to construct such a route. Does this mean that we have to search through every possible combination of flights and airlines? That is, do we need to ask every airline provider to communicate their entire schedule of flights to us? Are there techniques that we can employ to reduce the amount of flights we have to search through?

The amount of communication necessary to solve such problems is the primary objective of communication complexity. By studying communication complexity, we find lower and upper bounds on the minimum amount of communication necessary to solve such problems. In consequence, these bounds can inform on whether certain algorithms are optimal, or if it is possible to optimize them further. These bounds are increasingly relevant in a world full of distributed systems and data.

1.2 Communication Complexity Basics

Communication complexity studies the communication requirements to solve problems where inputs are distributed among at least 2 parties each with unbounded computing power. An example 2 party problem is typically stated as follows:

> There are two players, Alice and Bob, who respectively have a n-bit string, a and b, that may or may not be the same. The goal is to compute a function, $f(a, b)$, that depends on both a and b.

Under the two party model, it is always possible to compute $f(a, b)$ by sending all of a to b, since we assume each party has unbounded computing power. This approach of sending the entire input from one party to another is what is called the *trivial solution*. We say problems are *maximally hard* or *maximally complicated* when we cannot do asymptotically better than the *trivial solution*.

We call such problems maximally hard because it has a deterministic communication complexity of $\Theta(n)$ bits. We call it *deterministic* as we do not allow error. Thus in maximally hard problems, Alice cannot do asymptotically better than sending the n bits of a to Bob and the same for Bob. An example of a maximally hard problem is 2-player equality (EQ):

Claim 1. *Let $a, b \in \{0, 1\}^n$. That is, a and b are n-bit strings. Define equality as*

$$\mathsf{EQ}(a, b) = \begin{cases} 1 & a = b, \\ 0 & a \neq b. \end{cases} \tag{1}$$

The deterministic communication complexity of two party equality, denoted $D(\mathsf{EQ})$, is $\Omega(n)$.

Proof. We can represent the universe of all possible n bit inputs to the two parties using a $2^n \times 2^n$ boolean matrix M indexed by a and b. Then, let $M_{a,b} = 1$ if $a = b$ and $M_{a,b} = 0$ otherwise.

Therefore, M is the identity matrix. It is a well-known fact that the deterministic communication complexity of a problem is at least the $\log_2$ rank of its boolean matrix (Rao and Yehudayoff 2020). Therefore, since equality's matrix M is the $2^n \times 2^n$ identity matrix with rank 2^n, we have $D(\mathsf{EQ}) = \Omega(\log_2 2^n) = \Omega(n)$. □

$$
M_{b,a} = \begin{array}{c|cccc}
b\backslash a & 00 & 01 & 10 & 11 \\
\hline
00 & 1 & 0 & 0 & 0 \\
01 & 0 & 1 & 0 & 0 \\
10 & 0 & 0 & 1 & 0 \\
11 & 0 & 0 & 0 & 1
\end{array}
$$

Figure 1: Communication matrix for the equality function on two-bit inputs. The entry at column a and row b is 1 when $a = b$ and 0 otherwise.

As we will soon see, problems like equality are surprisingly useful for proving communication lower bounds for other, more complicated problems. That is, we say problem A is *at least as hard* as problem B if the communication complexity of A is at least the communication complexity of B.

1.3 The Graph Traversal Problem

In this report, we study the problem of finding a path in a graph which is jointly owned by various players. This is an abstract version of the motivating problem in 1.1. We choose to work in this abstract version so that that any bounds we prove will be applicable to a wide range of problems.

Informally, the graph traversal problem presents a graph, where players can own vertices. There is a start and end vertex. The player wish to know whether there exists a path from the start to the end through only vertices and edges owned by some player. Though each player knows what the graph looks like, they do not know who owns vertices and edges besides their own.

Formally, the graph traversal problem TRAV_2 is defined as follows:

Definition 1. *Let $G = (V, E)$ be an unweighted, undirected graph, where V is the set of all nodes and E is the set of edges. Alice and Bob each own a vertex subset $V_i \subseteq V$ of the graph, as well as all edges $E_i = \{(u, v) \in E \mid \{u, v\} \cap V_i \neq \emptyset\}$ connected to these vertices, for $i \in \{A, B\}$ for Alice and Bob respectively. Thus each player can be said owns a subgraph $G_i = (V_i, E_i)$ of G.*

Note that multiple players can own the same vertices and edges, and some vertices and edges may have no owner whatsoever.

Therefore, player inputs are solely determined by the vertices they own. Let the graph G have $|V| = n$ vertices. Then each player's subgraph can be represented by n-bit strings x and y, where 1 at index j indicates the player owns vertex j, and 0 means the player does not own j. We may use x, y and G_A, G_B interchangeably, as they encode the same information. Given a fixed G, start vertex v, and end vertex w, an instance of TRAV_2 *is defined as follows:*

$$\mathsf{TRAV}_2(x,y) : \{0,1\}^n \times \{0,1\}^n \to \{0,1\} \tag{2}$$

$$\mathsf{TRAV}_2(x,y) = \begin{cases} 1 & \exists \textit{ path from } v \textit{ to } w \textit{ through only vertices and edges in } G_A \cup G_B. \\ 0 & \textit{There is no such path.} \end{cases} \tag{3}$$

The above definition generalizes to TRAV_k, the traversal problem with k players, as seen in Definition 4. The following figures illustrate what TRAV_3 may look like.

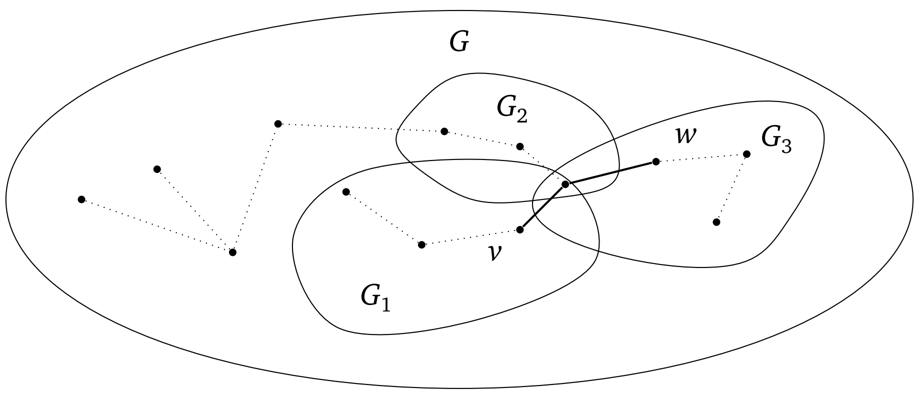

Figure 2: Example graph where $\mathsf{TRAV}(x,y) = 1$

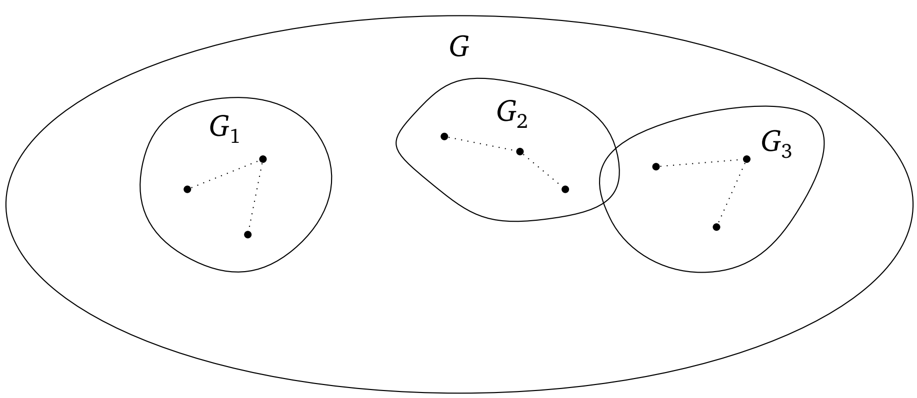

Figure 3: Example where $\mathsf{TRAV}(x,y) = 0$

2 The Two-Party Traversal Problem

2.1 Deterministic Two-party Bounds

We begin by considering the case with only two players and thus two subgraphs ($k = 2$). Let these players be named Alice and Bob, with subgraphs G_A and G_B respectively. Denote this problem by TRAV_2, i.e. the traversal problem for 2 players, where $|V(G)| = n$ and $|E(G)| = O(n^2)$. We claim that TRAV_2 is at least as hard as two player equality:

Claim 2. *Recall that equality* EQ *has inputs* $x, y \in \{0,1\}^n$*, and* $\mathsf{EQ}(x, y) = 1$ *iff* $x = y$*. Then* $D(\mathsf{TRAV}_2) \geq D(\mathsf{EQ})$*, and more generally* TRAV_2 *is at least as as hard as* EQ*.*

Proof. The high-level idea of this proof is to construct an equality gadget that lets us embed EQ into TRAV_2. That is, given x and y as inputs of EQ, we can convert these into subgraphs G_1 and G_2 inputs of TRAV. We convert this in such a way that $\mathsf{TRAV}(G_1, G_2) = 1$ if and only if $\mathsf{EQ}(x, y) = 1$. Then TRAV cannot use less communication than the optimal EQ protocol, else we get a contradiction.

Equality gadget. We construct a gadget that enforces equality at a single bit position i of x and y. The gadget is shown in Figure 4, and consists of two parallel branches:

1. The top branch corresponds to the assignment $x_i = y_i = 1$,
2. The bottom branch corresponds to the assignment $x_i = y_i = 0$.

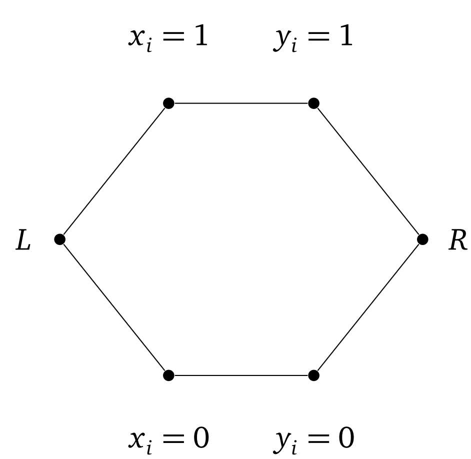

Figure 4: Gadget for proving equality. A path only exists if $x_i = y_i$ for either branch.

Add L and R into G_A and G_B. Then, add $x_i = 1$ if this is true to G_A, else add $x_i = 0$. Do the same for Bob. Then a path from L to R exists if and only if Alice and Bob have vertices on the same branch, which holds exactly when $x_i = y_i$.

Reduction to our graph problem. Construct a graph of n equality gadgets in a chain, as in Figure 5. Construct subgraph G_A as follows: Include all vertices at the start and end of each equality gadget (i.e. all the L and R above). Also, for every $i \in [n]$, give the vertex labelled $x_i = 1$ if this is true to G_A, else $x_i = 0$. Repeat for G_B.

In the resulting graph, set the leftmost vertex as the start vertex v, and the rightmost vertex as the end vertex w. Then there exists a path from v to w if and only if all gadgets are traversable, which occurs if and only if $x_i = y_i$ for all i, and therefore if and only if $x = y$.

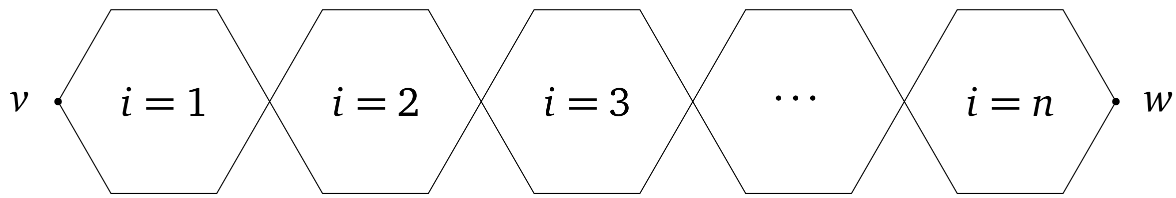

Figure 5: n equality gadgets chained together.

We conclude by noting that the resulting graph G has $\Theta(n)$ vertices and $\Theta(n)$ edges. As $|x| = |y| = \Theta(n)$, this completes the reduction and TRAV_2 is at least as hard as EQ. □

The trivial protocol where Alice sends all of G_A to Bob has communication complexity $\Theta(n)$. This gives $D(\mathsf{TRAV}_2) = O(n)$. Together with Claim 2, this gives $D(\mathsf{TRAV}_2) = \Theta(n)$. Thus TRAV_2 is deterministically maximally hard.

2.2 Randomized Two-Party Bounds

Above, we note that the problem is *deterministically maximally hard*. However, allowing for some error may allow us to produce a randomized algorithm which much better cost.

Consider the *public coin randomness* model. In this model, both parties have access to the same random bit-string in addition to the problem description and input. This random bit string can be as long as we want. We also allow our output to error, that is, be incorrect, some bounded constant percent of the time, and strictly less than half the time. It can be shown that using the public coin randomness model, equality—which is Deterministically maximally hard—can be reduced to being $O(1)$. We denote this by $R(\mathsf{EQ}) = O(1)$.

Unfortunately, for TRAV_2 we find that such a bound is unattainable. To prove this, we introduce the disjointness problem DISJ, which has public coin randomized communication complexity $R(\mathsf{DISJ}) = \Theta(n)$. As we show, DISJ can be embedded into TRAV_2.

Claim 3. *Define the disjointness problem as follows: Let $x, y \in \{0,1\}^n$ be n-bit strings representing subsets of $[n]$. Then, for all $i \in [n]$, if $x_i = 1$, we say i is in Alice's subset $S_A \subseteq [n]$, and if $x_i = 0$ it is not. Similarly for y and Bob. Then*

$$\mathsf{DISJ}(x,y) : \{0,1\}^n \times \{0,1\}^n \to \{0,1\} \tag{4}$$

$$\mathsf{DISJ}(x,y) = \begin{cases} 1 & S_A \cap S_B = \emptyset, \\ 0 & S_A \cap S_B \neq \emptyset. \end{cases} \tag{5}$$

It is well-known that $D(\mathsf{DISJ}) = \Theta(n)$ and $R(\mathsf{DISJ}) = \Theta(n)$ (Rao and Yehudayoff 2020). We claim $R(\mathsf{TRAV}_2) \geq R(\mathsf{DISJ})$, and more generally TRAV_2 is at least as hard as DISJ.

Proof. **Disjointness gadget.** We construct a gadget that enforces disjointness at a single entry $i \in [n]$. The gadget is shown in Figure 6. There, let x_i and y_i be indicator functions whether $i \in S_A$ and $i \in S_B$ respectively. Then:

1. The top branch corresponds to when $i \in S_A$ but $i \notin S_B$.
2. The middle branch corresponds to when $i \notin S_A$ and $i \notin S_B$.
3. The bottom branch corresponds to when $i \in S_B$ but $i \notin S_A$.

Give L and R to G_A and G_B. Then give all vertices labelled $x_i = 1$ to G_A if this is true, else give all vertices labelled $x_i = 0$. Do the same with y and G_B. In other words, there is no path L to R if $i \in X \cap Y$, but there is a path in all other cases.

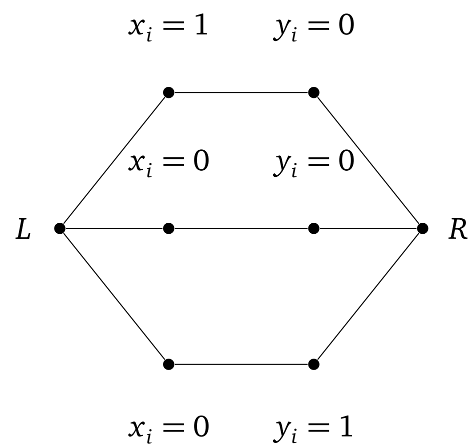

Figure 6: Disjoitness gadget. A path L to R only exists if either $x_i = 0$ or $y_i = 0$.

Reduction to our graph problem. We construct a graph G by chaining one copy of this gadget for each index $i \in [n]$ between a global start vertex v and end vertex w. Construct G_A by giving all vertices at the start and end of every disjointness gadget (i.e. all L and R), and vertices labelled $x_i = 1$ if it is true, else $x_i = 0$. Do similar for G_B.

In the resulting graph, there exists a path from v to w if and only if all disjointness gadgets are traversable, which occurs if and only if $X \cap Y = \emptyset$.

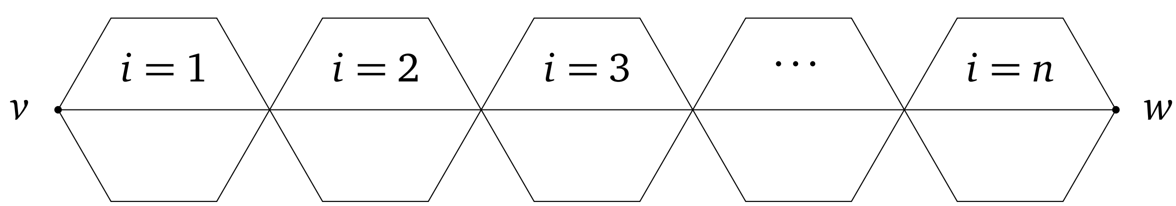

Figure 7: Disjointness gadgets chained together.

We conclude by noting that $|V(G)| = \Theta(n)$ and $|E(G)| = \Theta(n)$. As $|x| = |y| = \Theta(n)$, this completes the reduction and TRAV_2 is at least as hard as DISJ. $\square$

Recall that in Claim 2, we showed that $D(\mathsf{TRAV}_2) = \Theta(n)$. With disjointness, we show that

$R(\mathsf{TRAV}_2) = \Theta(n)$ too. This is because Claim 3 gives $R(\mathsf{TRAV}_2) = \Omega(n)$, since $R(\mathsf{DISJ}) = \Theta(n)$ (Rao and Yehudayoff 2020). Then, the trivial protocol where Alice sends her entire G_A to Bob in $\Theta(n)$ bits works even in the public coin randomness model. Together this gives $R(\mathsf{TRAV}_2) = \Theta(n)$. In other words, we cannot do much better even with bounded error.

2.3 Specialized Two-Party Bounds

The previous proofs show that TRAV_2 is maximally complicated in terms of deterministic communication complexity and public randomized communication complexity. A natural follow up question is to ask if we can do better. That is, given some additional constraints, can we solve the problem with less communication asymptotically?

Definition 2. *Define the CTRAV_2 problem as a TRAV_2 problem where G_A and G_B are guaranteed to be complete and connected subgraphs of G.*

Claim 4. *CTRAV_2 is at least as hard as DISJ. Thus, $D(\mathsf{CTRAV}_2) = \Theta(n)$ and $R(\mathsf{CTRAV}_2) = \Theta(n)$.*

Proof. If the start v or end w is in neither G_A nor G_B or both in one of G_A or G_B, the problem can be resolved in $O(1)$ communication. So suppose $v \in G_A$ and $w \in G_B$.

Since G_A and G_B are complete, a valid path occurs if and only if $V(G_A) \cap V(G_B) \neq \emptyset$. Then, by a similar argument to Claim 3, this problem is at least as hard as DISJ. More explicitly, let x, y be inputs to DISJ. We convert x, y into subgraphs G_A, G_B of a complete graph G with n vertices. This conversion is by including vertex i in G_A if $x_i = 1$, and not including it if $x_i = 0$. Do the same fot G_B and y.

If $\mathsf{CTRAV}_2(G_A, G_B) = 1$, then $V(G_A) \cap V(G_B) \neq \emptyset$. So $x \cap y \neq \emptyset$ and $\mathsf{DISJ}(x, y) = 0$. Similarly, if $\mathsf{CTRAV}_2(G_A, G_B) = 0$, then $V(G_A) \cap V(G_B) = \emptyset$. Thus $x \cap y = \emptyset$ and $\mathsf{DISJ}(x, y) = 1$. We note that $|V(G_A)| = |V(G_B)| = |x| = |y| = \Theta(n)$. This completes the reduction. □

Informally, this tells us that the local structure of G_A and G_B alone is not a significant source of communication difficulty. That is, even when G_A and G_B are complete graphs, we do not immediately get better bounds and the problem does not become significantly easier. Since G_A and G_B are complete, we can say that these subgraphs have the maximum number of edges. We now consider a sparser case:

Definition 3. *Define STRAV_2 as a TRAV_2 problem where $E(G) = O(n)$, $E(G_A) = O(n)$, and $E(G_B) = O(n)$. In other words, the graph G is sparse and G_A, G_B are sparse too.*

Claim 5. *STRAV_2 is at least as hard as DISJ. Thus, $D(\mathsf{STRAV}_2) = \Theta(n)$ and $R(\mathsf{STRAV}_2) = \Theta(n)$.*

Proof. We reuse the reduction from DISJ to TRAV_2 in Claim 3. In that construction, G is a chain of n disjointness gadgets, so $|E(G)| = \Theta(n)$. Moreover, each player's subgraph consists of the gadget boundary vertices and one branch choice per gadget, which also gives $|E(G_A)| = \Theta(n)$ and $|E(G_B)| = \Theta(n)$.

Hence every instance produced by that reduction already satisfies the sparsity constraints in the definition of STRAV_2. The reduction still preserves correctness: $\mathsf{DISJ}(x, y) = 1$ iff there is a v-w path in $G_A \cup G_B$. Therefore STRAV_2 is at least as hard as DISJ. □

Informally, this tells us that weak edge sparsity of G, G_A, and G_B does not add nor remove significant communication cost. This may lead us to believe that the local structure of G, G_A, and G_B, namely, how sparse or dense the edges corresponding to the owned vertices are, do not have a significant impact on the deterministic and pubic randomized communication complexity of the TRAV_2 problem.

3 The Multi-Party Traversal Problem

In Section 2, we analyzed TRAV_2, the two-party traversal problem. We now study the k-party version, denoted TRAV_k, with $k \geq 2$.

Definition 4. *Let $G = (V, E)$ be an undirected graph with designated vertices $v, w \in V$, where $|V| = n$. There are k players, and player $i \in [k]$ owns a vertex subset $V_i \subseteq V$. The edge set owned player i is $E_i = \{(a, b) \in E \mid \{u, v\} \cap V_i \neq \emptyset\}$. Thus player i owns a subgraph $G_i = (V_i, E_i)$. Given a fixed G, start vertex v, and end vertex w known to all players, an instance of TRAV_k is defined as follows:*

$$\mathsf{TRAV}_k(G_1, \ldots, G_k) : \bigtimes_{i=1}^{k} \{0,1\}^n \to \{0,1\} \tag{6}$$

$$\mathsf{TRAV}_k(G_1, \ldots, G_k) = \begin{cases} 1 & \textit{if there is a path from } v \textit{ to } w \textit{ in } \bigcup_{i=1}^{k} G_i, \\ 0 & \textit{otherwise.} \end{cases} \tag{7}$$

Definition 5. *The k-party disjointness problem DISJ_k is defined as follows: Each player $i \in [k]$ has input $x^{(i)} \in \{0,1\}^n$, interpreted as a subset $S_i = \{j \in [n] \mid x_j^{(i)} = 1\} \subseteq [n]$. Define*

$$\mathsf{DISJ}_k(x^{(1)}, \ldots, x^{(k)}) : \bigtimes_{i=1}^{k} \{0,1\}^n \to \{0,1\} \tag{8}$$

$$\mathsf{DISJ}_k(x^{(1)}, \ldots, x^{(k)}) = \begin{cases} 1 & \bigcap_{i=1}^{k} S_i = \emptyset, \\ 0 & \bigcap_{i=1}^{k} S_i \neq \emptyset. \end{cases} \tag{9}$$

It is well known that $D(\mathsf{DISJ}_k) = \Omega(nk)$ and $R(\mathsf{DISJ}_k) = \Omega(nk)$ (Braverman et al. 2013).

Claim 6. *TRAV_k is at least as hard as DISJ_k. Furthermore, $D(\mathsf{TRAV}_k) = \Theta(nk)$ and $R(\mathsf{TRAV}_k) = \Theta(nk)$.*

Proof. Let $(x^{(1)}, \ldots, x^{(k)})$ be an input to DISJ_k with $x^{(i)} \in \{0,1\}^n$. The high level idea is similar to Claim 3: we build a TRAV_k instance from a chain of n multi-party disjointness gadgets.

Multi-Party Disjointness gadget. For each index $j \in [n]$, create vertices L_j, R_j, and for each player $i \in [k]$, a vertex $u_{i,j}$. All players own L_j and R_j. Furthermore, $u_{i,j} \in V_i$ if and only if $x_j^{(i)} = 0$. Then there is a path from L_j to R_j if and only if at least one player has $x_j^{(i)} = 0$. Equivalently, there is no such path if and only if all players have $x_j^{(i)} = 1$.

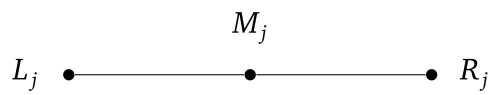

Figure 8: Gadget for embedding DISJ_k. A valid path L_j to R_j exists if and only if $\exists i \in [k]$ with $x_j^{(i)} = 0$.

Reduction to TRAV_k. We construct a graph G by chaining n copies of the multi-party disjointness gadget, connecting $R_l = L_{l+1}$ for all $l \in [n-1]$. Set the start vertex $v = L_1$ and end vertex $w = R_n$. Then there is a path from v to w in $\bigcup_{i=1}^{k} G_i$ if and only if every gadget is traversable (that is, there is a path L_j to R_j for all $j \in [n]$). This means that for every j, there is some player with $x_j^{(i)} = 0$, so $\bigcap_{i=1}^{k} S_i = \emptyset$. In other words, a path from v to w witnesses that every element in $[n]$ is missing from at least one subset S_i.

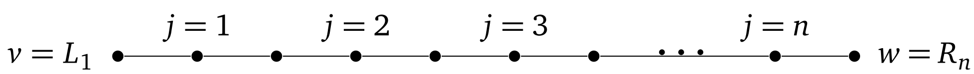

Figure 9: n multi-party disjointness gadgets chained together.

We conclude the proof by noting that the construction uses $\Theta(n)$ vertices and edges each. Hence $D(\mathsf{TRAV}_k) = \Omega(nk)$ and $R(\mathsf{TRAV}_k) = \Omega(nk)$.

For upper bounds, consider the trivial protocol where players $2, \ldots, k$ send their input to player 1 in $(k-1)n = \Theta(nk)$ bits. Then Player 1 can compute the answer. This trivial protocol works in both the deterministic and randomized setting. This gives $D(\mathsf{TRAV}_k) = \Theta(nk)$ and $R(\mathsf{TRAV}_k) = \Theta(nk)$. $\square$

Informally, this tells us that adding more players does not make traversal an easier problem in terms of deterministic and public randomized communication complexity.

4 Closing Remarks

In this paper, we explored the communication complexity of multiparty graph traversal as defined in the TRAV problem and its variants. By exploring embeddings of equality an disjointness, we show TRAV is maximally hard under a variety of conditions in the deterministic and public randomized communication complexity models. In other words, given the most general TRAV_k for $k \geq 2$, one cannot asymptotically do better than the trivial solution of sending all inputs to one player.

This has a variety of implications for algorithm, system, and database designers. If such specialists were to implement algorithms solving TRAV_k in it its most general form, we have proven that there are no "communication" tricks that can be used to get or even approximate the right answer. More formally, there does not exist an algorithm that computes TRAV_k asymptotically better than the trivial protocol that attains at most a constant bounded error against every possible input.

In particular, we note that the trivial protocol can be implemented by having one player act as the "coordinator," receiving and processing everyone's inputs. In terms of minimizing bits communicated, this means systems that solve problems like TRAV_k may benefit from having a single, optimized server as opposed to distributed computing. Of course, there are practical, economical, and privacy reasons that may complicate the choice of implementation and the priority of minimizing communication in practice.

4.1 Future Work

Though we have shown that TRAV_k and some of its variants are maximally hard under the deterministic and randomized model, it is entirely possible that specialized instances of TRAV_k give rise to clever algorithms and lower bounds that have significantly less communication complexity. Below, we propose some such variants, which due to practical constraints we cannot examine in this paper.

1. **Bounded-length path**. Consider a TRAV_2 problem where we know that if there is a path from v to w in G, then this path is of length at most $l = O(1)$. Does this give rise to a significantly lower communication complexity? What about for TRAV_k?
2. **Single-connected component**. We noted that CTRAV_2 is also maximally complicated in the deterministic and public coin model. Notice that G_A and G_B in CTRAV_2 are connected components. If we guarantee in TRAV_k that each G_i is a single connected component, can we achieve lower communication complexity?
3. **Bounded-owners**. Suppose it is known that each vertex in TRAV_k can be owned by at most $p = O(1)$ players. For example, if $p = 1$, each vertex can be owned by at most 1 player. Does this give rise to improved communication lower and upper bounds?

It may also be worthwhile to investigate very specific instances of TRAV based on real-world networks and graphs. For example, perhaps a combination of the above variants models a specific, real-world problem that gives rise to clever algorithms with non-maximal communication complexity. That said, one benefit of studying the most general cases is the ability for the results to be applicable to a wide range of problems, scenarios, and applications.

5 References

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